Tor Nelson

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Summary

Automation & Systems Engineer with 5+ years of experience designing production automation, internal tooling, and reliability-focused workflows across distributed environments. Specialized in platform automation, self-healing systems, and operational efficiency through software-driven solutions. Focused on software engineering work spanning automation, CI/CD, and production systems.

Work Experience

Automation Engineer / Systems Engineer (Internal Tooling & Platform Automation)
Prairie IT Services | 5 / 2019 - Current

Built and operated internal automation platforms and reliability tooling supporting managed production environments. Developed solutions to reduce manual toil, standardize system behavior, and introduce software-driven control systems for monitoring and remediation.

Key Accomplishments:

Automation & Internal Tooling

- Architected and maintained an internal PowerShell automation library of reusable components, standardizing operations and reducing manual intervention.
- Designed composable function frameworks for Exchange Online and Microsoft 365 administration, replacing point-and-click workflows with version-controlled automation.
- Built automated ticket-generation and workflow orchestration systems integrating monitoring data with service management pipelines, improving operational throughput and predictability.

Reliability, Monitoring & Self-Healing Systems

- Designed automated monitoring and remediation pipelines to detect system instability and trigger corrective workflows, reducing unplanned downtime and frontline support escalations.
- Implemented self-healing automation for system file corruption detection and repair, improving system stability and decreasing recurring incident volume.
- Developed failure-detection pipelines attaching structured diagnostic data to incident tickets, accelerating root-cause analysis and resolution.

Production Operations & Control Systems

- Built automation to enforce operational contracts around backup reliability, improving success rates by 20% through detection, validation, and automated recovery workflows.
- Designed repeatable control processes for cybersecurity audits and system health checks, formalizing operations with automation, internal tooling, and documentation.

AI Literacy & Prompt Optimization Platform — Founder & Product Engineer | 2025–Present

Designed and built an AI literacy and prompt optimization platform that helps users systematically improve the quality, reliability, and usefulness of AI outputs using composable optimization strategies.

- Architected and implemented a full-stack web application using MongoDB, Express, React, and Node.js.
- Designed a modular prompt optimization engine encoding proven communication and reasoning patterns with a focus on creating an experiential learning journey to develop AI literacy skills.
- Built a mature automated test suite spanning unit, integration, and end-to-end tests, using AI-assisted test generation with manual review to enforce behavioral contracts and prevent regressions.
- Applied specification-first and test-driven development practices, treating tests as executable contracts governing system behavior, reliability, and user-facing expectations.
- Developed and validated a novel AI-first, spec-driven development workflow that accelerates software delivery while preserving control boundaries, correctness, and system integrity.
- Currently preparing the platform for public launch with a focus on production readiness, security hardening, accessibility, and user experience.

Skills

- **Programming:**
    - Languages: JavaScript, Python, PowerShell
    - Frameworks & Stacks: Node.js, Express, React, MongoDB (MERN)
    - Automation & Scripting: PowerShell, Python, Bash
    - Testing & Quality: Automated Testing, AI-Assisted Test Generation, Systematic Program Design

- **Production Systems & Automation:**
    - Self-Healing Systems & Remediation Pipelines
    - Internal Tooling & Platform Automation
    - Monitoring & Observability
    - CI/CD principles and test-driven release workflows
    - Reliability Engineering Foundations
- **Data Analysis & Problem Solving:**
    - Analytical problem-solver with a strong background in mathematics, data analysis, and statistical modeling. Proven track record of identifying issues and finding creative solutions to improve operational efficiency and reliability.
    - Experience devising and conducting scientific experiments as part of a research and development program.
    - Familiar with design thinking and user experience design methodologies to inform business and product development.
- **Soft Skills:**
    - Exceptional communication skills. Able to explain complex technical information to non-technical audiences.
    - Continuous learner and skilled practitioner of strengths-based coaching and knowledge management.

Education

Bachelor of Science
The Evergreen State College

Professional Development:
Completed individual courses in Computer Science, Data Science, Mathematics, Design, Business, and Psychology through Coursera, Ed-X, Udemy, Pluralsight, and other sources.